

MASTER-KIDS · OLYMPIAD PRACTICE SERIES · CLASS 4

Mathematics Olympiad Worksheet 33

Level 2 — Intermediate · Measurement II

Class	Class 4 (Primary · Stage 2)	Subject	Mathematics
Level	Level 2 — Intermediate	Topic Focus	Measurement II — Multi-Step Word Problems
Questions	20	Time Allowed	35 minutes
Total Marks	20	Worksheet No.	33 of 100
Curriculum	CBSE · ICSE · State Boards (board-agnostic) · NEP 2020-aligned		

About Master-Kids

Master-Kids is India's cradle-to-career learning platform — one platform that grows with a child from Pre-Nursery all the way to college, turning everyday effort into visible progress for parents and a rewarding game of streaks, XP and badges for children. This Olympiad practice series is part of the Master-Kids Primary School programme (Stage 2, Class 1-5). It is board-agnostic — suited to CBSE, ICSE and State Board children alike — and aligned with the NEP 2020 focus on foundational numeracy, science thinking and language.

About This Worksheet

Worksheet 33 is pure applied measurement — trucks with weight limits, airline baggage allowance, petrol mileage, mixing and repacking, staircases and buried poles. Nearly every question takes two or three steps (convert → compute → convert back), exactly the Everyday Mathematics style of the IMO at intermediate difficulty.

Learning Objectives

- Solve two- and three-step problems across length, weight and capacity.
- Work with limits and allowances (how much more can be added).
- Apply rate reasoning: petrol per km, cost per metre, dose per day.
- Mix, repack and share quantities using all four operations.
- Convert between units fluently inside word problems.

Instructions

1. This worksheet has 20 multiple-choice questions. Each question has exactly one correct answer.
2. Read every question carefully. Look at any picture, table or figure before choosing.
3. Each correct answer earns +1 mark. There is no negative marking, so attempt every question.
4. Write your chosen option — (A), (B), (C) or (D) — clearly in the answer box or on a separate sheet.
5. Do all rough work in the margin or on rough paper. Do not use a calculator.
6. After finishing, check your answers against the Answer Key, then read the Detailed Solutions to understand the reasoning behind every question — that is where the real learning happens.

Marking scheme: +1 mark for each correct answer · No negative marking · Maximum 20 marks.

**SECTION A · Multiple Choice Questions · 20 Questions**

Q1. A drum holds 45 L of water. It is filled 3 times in a day and emptied completely each time for cleaning a hall. How much water is used in the day?

- (A) 90 L (B) 135 L
(C) 120 L (D) 145 L

Q2. A tailor needs 1 m 75 cm of cloth for one shirt. How much cloth is needed for 4 shirts?

- (A) 6 m (B) 7 m
(C) 7 m 50 cm (D) 6 m 75 cm

Q3. A truck can carry 5,000 kg. It is loaded with 35 sacks of 120 kg each. How much MORE weight can it carry?

- (A) 800 kg (B) 900 kg
(C) 1,000 kg (D) 700 kg

Q4. Riya mixes 750 mL of mango pulp with 1 L 250 mL of milk to make a shake. She pours it equally into 4 glasses. How much shake is in each glass?

- (A) 400 mL (B) 500 mL
(C) 600 mL (D) 450 mL

Q5. A ribbon of 7 m 20 cm is cut into pieces of 90 cm each. How many pieces are made?

- (A) 7 (B) 8
(C) 9 (D) 10

Q6. A bucket holds 9 L 500 mL. A tank needs 38 L of water. How many FULL buckets are needed to have at least 38 L?

- (A) 3 (B) 4
(C) 5 (D) 6

Q7. Aman walks 850 m to the park, jogs 1 km 350 m inside it, and walks 850 m back. What TOTAL distance does he cover?

- (A) 2 km 850 m (B) 3 km 50 m
(C) 3 km (D) 2 km 950 m

Q8. A grocer mixes 2 kg 250 g of almonds with 1 kg 750 g of raisins, then packs the mixture into packets of 500 g. How many packets does he make?

- (A) 6 (B) 7
(C) 8 (D) 9

Q9. A pole is 6 m long. 1 m 20 cm of it is buried in the ground. What length is ABOVE the ground?

- (A) 4 m 80 cm (B) 5 m 20 cm
(C) 4 m 20 cm (D) 5 m 80 cm



- Q10.** A car uses 1 L of petrol for every 15 km. How much petrol is needed for a 90 km trip?
- (A) 5 L (B) 6 L
(C) 7 L (D) 9 L
- Q11.** A baby elephant drinks 11 L of milk each day. How much milk does it drink in a WEEK?
- (A) 66 L (B) 70 L
(C) 77 L (D) 84 L
- Q12.** 3 m of wire costs ₹84. What does 5 m of the same wire cost?
- (A) ₹120 (B) ₹140
(C) ₹160 (D) ₹130
- Q13.** A jug of 2 L is full. 3 glasses of 250 mL each are poured out. How much water REMAINS in the jug?
- (A) 1 L 250 mL (B) 1 L 500 mL
(C) 750 mL (D) 1 L 750 mL
- Q14.** One brick is 20 cm long. Bricks are laid end to end along a 5 m wall. How many bricks fit in one row?
- (A) 20 (B) 25
(C) 30 (D) 40
- Q15.** Sara's suitcase weighs 18 kg 500 g. The airline limit is 23 kg. How much MORE can she pack?
- (A) 4 kg 500 g (B) 5 kg 500 g
(C) 4 kg (D) 5 kg
- Q16.** A farmer waters his field with 6 tankers of 1,500 L each. He needs 10,000 L in all. How much MORE water is required?
- (A) 500 L (B) 1,000 L
(C) 1,500 L (D) 2,000 L
- Q17.** A staircase has 20 steps, each 15 cm high. How HIGH does the staircase rise in metres?
- (A) 2 m (B) 2 m 50 cm
(C) 3 m (D) 3 m 50 cm
- Q18.** A milk booth sells 48 L of milk in the morning and 36 L 500 mL in the evening. How much milk is sold in the day?
- (A) 84 L 500 mL (B) 85 L 500 mL
(C) 84 L (D) 83 L 500 mL
- Q19.** A rope is cut into 6 equal pieces of 1 m 25 cm each. What was the ORIGINAL length of the rope?
- (A) 6 m 50 cm (B) 7 m
(C) 7 m 50 cm (D) 8 m



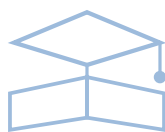
Q20. A recipe needs 1 kg 200 g of flour. Chef Anu has only 850 g. She borrows the REST from a neighbour. How much does she borrow?

(A) 250 g

(B) 350 g

(C) 450 g

(D) 550 g





ANSWER KEY & DETAILED SOLUTIONS

Quick Answer Key

Q	1	2	3	4	5	6	7	8	9	10
Ans	B	B	A	B	B	B	B	C	A	B
Q	11	12	13	14	15	16	17	18	19	20
Ans	C	B	A	B	A	B	C	A	C	B

Step-by-Step Solutions

Q1. Correct answer: (B)

$$3 \times 45 = 135 \text{ L.}$$

Q2. Correct answer: (B)

$$175 \text{ cm} \times 4 = 700 \text{ cm} = 7 \text{ m.}$$

**Q3. Correct answer: (A)**

$$\text{Loaded} = 35 \times 120 = 4,200 \text{ kg. Spare} = 5,000 - 4,200 = 800 \text{ kg.}$$

Q4. Correct answer: (B)

$$\text{Total} = 750 + 1,250 = 2,000 \text{ mL. Each glass} = 2,000 \div 4 = 500 \text{ mL.}$$

Q5. Correct answer: (B)

$$720 \div 90 = 8 \text{ pieces.}$$

Q6. Correct answer: (B)

$$4 \text{ buckets} = 4 \times 9,500 = 38,000 \text{ mL} = 38 \text{ L exactly} \rightarrow 4 \text{ buckets.}$$

Q7. Correct answer: (B)

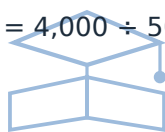
$$850 + 1,350 + 850 = 3,050 \text{ m} = 3 \text{ km } 50 \text{ m.}$$

Q8. Correct answer: (C)

$$\text{Mixture} = 2,250 + 1,750 = 4,000 \text{ g. Packets} = 4,000 \div 500 = 8.$$

Q9. Correct answer: (A)

$$600 - 120 = 480 \text{ cm} = 4 \text{ m } 80 \text{ cm.}$$

**Q10. Correct answer: (B)**

$$90 \div 15 = 6 \text{ L.}$$

Q11. Correct answer: (C)

$$11 \times 7 = 77 \text{ L.}$$

Q12. Correct answer: (B)

$$1 \text{ m} = 84 \div 3 = ₹28. 5 \text{ m} = 5 \times 28 = ₹140.$$

Q13. Correct answer: (A)

$$\text{Poured} = 3 \times 250 = 750 \text{ mL. Remains} = 2,000 - 750 = 1,250 \text{ mL} = 1 \text{ L } 250 \text{ mL.}$$

Q14. Correct answer: (B)

$$5 \text{ m} = 500 \text{ cm. } 500 \div 20 = 25 \text{ bricks.}$$

Q15. Correct answer: (A)

$$23,000 - 18,500 = 4,500 \text{ g} = 4 \text{ kg } 500 \text{ g.}$$



Q16. Correct answer: (B)

Delivered = $6 \times 1,500 = 9,000$ L. More needed = $10,000 - 9,000 = \mathbf{1,000}$ L.

Q17. Correct answer: (C)

$20 \times 15 = 300$ cm = **3 m**.

Q18. Correct answer: (A)

$48,000 + 36,500 = 84,500$ mL = **84 L 500 mL**.

Q19. Correct answer: (C)

$125 \times 6 = 750$ cm = **7 m 50 cm**.

Q20. Correct answer: (B)

$1,200 - 850 = \mathbf{350}$ g.

